From: Jeff Dietlin <utilities@Cadillac-MI.net>
Sent: 10/29/2018 9:56:58 AM
To: "Berdinski, Thomas (DEQ)" <BERDINSKIT@michigan.gov>
Cc: "Doug Langworthy" <dlangworthy@Cadillac-MI.net>; "Cindy Tomaszewski" <ctomaszewski@Cadillac-MI.net>
Subject: FW: TestAmerica invoice and report files from 190-17646-1 City of Cadillac
Attachments: J17646-1 UDS Level 2 Report Final Report.pdf, deq-tou-WRD-IPP_PFAS_Guidance-ScreeningEvaluation_620434_7.pdf

Tom,

Please see the attached. I assume since we now have these results we should test our effluent to see if it is making it through the plant. Is this correct?

Since this leachate is a trucked in source we could stop it at any time. We will be working with the landfill to try to get some treatment before it comes to the WWTP.

Should we test our influent to determine if there is any other sources we missed on our inventory? Or wait until the effluent test results come back?

Thank you,

Jeffrey M. Dietlin
City of Cadillac
Director of Utilities
(231) 779-7346(Direct)
(231)775-0906 (Fax)

From: Lab
Sent: Monday, October 29, 2018 9:51 AM
To: Jeff Dietlin
Subject: FW: TestAmerica invoice and report files from 190-17646-1 City of Cadillac

Jeff –

I'm forwarding the results of the leachate PFAS testing we received from TestAmerica, this morning. These results exceed the applicable water quality standards for PFOS (12 ng/L) . The criteria are given in "Table 1 – Applicable Rule 57 Values" of the IPP PFAS INITIATIVE SCREEENING EVALUATION GUIDANCE (attached).

I will order a set of bottles and schedule testing for our effluent. If you need anything else, just let me know. Thanks,

Cindy A. Tomaszewski
Lab Supervisor

From: Schafer, Sue [mailto:sue.schafer@testamericainc.com]
Sent: Monday, October 29, 2018 7:56 AM
To: Lab
Subject: TestAmerica invoice and report files from 190-17646-1 City of Cadillac

Hello,

Attached please find the invoice and report files for job 190-17646-1; City of Cadillac

Please feel free to contact me if you have any questions.

Thank you.

Please let us know if we met your expectations by rating the service you received from TestAmerica on this project by visiting our website at: [Project Feedback]

SUE SCHAFER
Project Manager

TestAmerica
THE LEADER IN ENVIRONMENTAL TESTING

Tel: 810.229.2763
www.testamericainc.com

Reference: [040710]
Attachments: 2

ATTACHMENT NAME:

J17646-1 UDS Level 2 Report Final Report.pdf

ATTACHMENT TYPE:

Adobe Portable Document Format (PDF) compound image

THE LEADER IN ENVIRONMENTAL TESTING

ANALYTICAL REPORT

TestAmerica Laboratories, Inc.
TestAmerica Michigan
10448 Citation Drive
Suite 200
Brighton, MI 48116
Tel: (810)229-2763

TestAmerica Job ID: 190-17646-1
Client Project/Site: City of Cadillac

For:
City of Cadillac Utilities
1121 Plett Road
Cadillac, Michigan 49601

Attn: Cindy Tomaszewski

Sue Schafer

Authorized for release by:
10/29/2018 7:51:04 AM

Sue Schafer, Project Manager II
(810)229-2763
sue.schafer@testamericainc.com

This report has been electronically signed and authorized by the signatory. Electronic signature is intended to be the legally binding equivalent of a traditionally handwritten signature.

Results relate only to the items tested and the sample(s) as received by the laboratory.

Table of Contents

Sample Summary

Client: City of Cadillac Utilities
Project/Site: City of Cadillac

TestAmerica Job ID: 190-17646-1

Lab Sample ID	Client Sample ID	Matrix	Collected	Received
190-17646-1	Leachate	Water	10/03/18 09:30	10/04/18 12:41
190-17646-2	Leachate (duplicate)	Water	10/03/18 09:35	10/04/18 12:41
190-17646-3	Equipment Blank	Water	10/03/18 09:25	10/04/18 12:41
190-17646-4	Field Blank	Water	10/03/18 00:00	10/04/18 12:41

Job ID: 190-17646-1

Laboratory: TestAmerica Michigan

Narrative

Job Narrative
190-17646-1

Comments
No additional comments.

Receipt
The samples were received on 10/4/2018 12:41 PM; the samples arrived in good condition, properly preserved and, where required, on ice. The temperature of the cooler at receipt was 0.0º C.

LCMS
Method(s) 537 (modified): Results for samples Leachate (190-17646-1) and Leachate (duplicate) (190-17646-2) were reported from the analysis of a diluted extract due to matrix interferences in the analysis of the undiluted extract. The dilution factor was applied to the labeled internal standard area counts and these area counts were within acceptance limits

Method(s) 537 (modified): Results for samples Leachate (190-17646-1) were reported from the analysis of a diluted extract due to matrix interferences in the analysis of the undiluted extract. The dilution factor was applied to the labeled internal standard area counts and these area counts were within acceptance limits.

Method(s) 537 (modified): The following samples were diluted due to the nature of the sample matrix: Leachate (190-17646-1) and Leachate (duplicate) (190-17646-2). Elevated reporting limits (RLs) are provided.

Method(s) 537 (modified): The Isotope Dilution Analyte (IDA) recovery for Perfluorotetradecanoic acid (PFTeA) associated with the following samples is below the method recommended limit: Leachate (190-17646-1) and Leachate (duplicate) (190-17646-2). This sample was re-analyzed at greater dilution with concurring results. Generally, data quality is not considered affected if the IDA signal-to-noise ratio is greater than 10:1, which is achieved for all IDA in the samples.

Method(s) 537 (modified): Results for samples Leachate (duplicate) (190-17646-2) were reported from the analysis of a diluted extract due to matrix interferences in the analysis of the undiluted extract. The dilution factor was applied to the labeled internal standard area counts and these area counts were outside acceptance limits. The internal standard is not used to quantitate target analytes; therefore, there is no impact to the data.

No additional analytical or quality issues were noted, other than those described above or in the Definitions/Glossary page.

Organic Prep
Method(s) 3535: The following samples are a brown color prior to extraction:
Leachate (190-17646-1) and Leachate (duplicate) (190-17646-2).

Method Code: 3535 PFC
preparation batch 320-252552

Method(s) 3535: The following samples are a yellow color after extraction:
Leachate (190-17646-1) and Leachate (duplicate) (190-17646-2).

Method Code: 3535 PFC
preparation batch 320-252552

No additional analytical or quality issues were noted, other than those described above or in the Definitions/Glossary page.

Client Sample Results

Client: City of Cadillac Utilities
Project/Site: City of Cadillac

TestAmerica Job ID: 190-17646-1

Client Sample ID: Leachate
Date Collected: 10/03/18 09:30
Date Received: 10/04/18 12:41

Lab Sample ID: 190-17646-1
Matrix: Water

Method: 537 (modified) - Fluorinated Alkyl Substances

Analyte	Result	Qualifier	RL		Unit	D	Prepared	Analyzed	Dil Fac
Perfluorobutanoic acid (PFBA)	**950**		21		ng/L		10/16/18 08:36	10/25/18 04:26	10
Perfluoropentanoic acid (PFPeA)	**610**		21		ng/L		10/16/18 08:36	10/25/18 04:26	10
Perfluorohexanoic acid (PFHxA)	**2100**		21		ng/L		10/16/18 08:36	10/25/18 04:26	10
Perfluoroheptanoic acid (PFHpA)	**240**		21		ng/L		10/16/18 08:36	10/25/18 04:26	10
Perfluorooctanoic acid (PFOA)	**590**		21		ng/L		10/16/18 08:36	10/25/18 04:26	10
Perfluorononanoic acid (PFNA)	<21		21		ng/L		10/16/18 08:36	10/25/18 04:26	10
Perfluorodecanoic acid (PFDA)	<21		21		ng/L		10/16/18 08:36	10/25/18 04:26	10
Perfluoroundecanoic acid (PFUnA)	<21		21		ng/L		10/16/18 08:36	10/25/18 04:26	10
Perfluorododecanoic acid (PFDoA)	<21		21		ng/L		10/16/18 08:36	10/25/18 04:26	10
Perfluorotridecanoic acid (PFTriA)	<21		21		ng/L		10/16/18 08:36	10/25/18 04:26	10
Perfluorotetradecanoic acid (PFTeA)	<21		21		ng/L		10/16/18 08:36	10/25/18 04:26	10
Perfluorobutanesulfonic acid (PFBS)	**950**		21		ng/L		10/16/18 08:36	10/25/18 04:26	10
Perfluoropentanesulfonic acid (PFPeS)	**160**		21		ng/L		10/16/18 08:36	10/25/18 04:26	10
Perfluorohexanesulfonic acid (PFHxS)	**610**		21		ng/L		10/16/18 08:36	10/25/18 04:26	10
Perfluoroheptanesulfonic Acid (PFHpS)	<21		21		ng/L		10/16/18 08:36	10/25/18 04:26	10
Perfluorooctanesulfonic acid (PFOS)	**120**		21		ng/L		10/16/18 08:36	10/25/18 04:26	10
Perfluorononanesulfonic acid (PFNS)	<21		21		ng/L		10/16/18 08:36	10/25/18 04:26	10
Perfluorodecanesulfonic acid (PFDS)	<21		21		ng/L		10/16/18 08:36	10/25/18 04:26	10
Perfluorooctanesulfonamide (FOSA)	<21		21		ng/L		10/16/18 08:36	10/25/18 04:26	10
N-methylperfluorooctanesulfonamidoacetic acid (NMeFOSAA)	<210		210		ng/L		10/16/18 08:36	10/25/18 04:26	10
N-ethylperfluorooctanesulfonamidoacetic acid (NEtFOSAA)	<210		210		ng/L		10/16/18 08:36	10/25/18 04:26	10
4:2 FTS	<210		210		ng/L		10/16/18 08:36	10/25/18 04:26	10
8:2 FTS	<210		210		ng/L		10/16/18 08:36	10/25/18 04:26	10

Isotope Dilution	*%Recovery*	*Qualifier*	*Limits*	*Prepared*	*Analyzed*	*Dil Fac*
13C4 PFBA	*27*		*25 - 150*	*10/16/18 08:36*	*10/25/18 04:26*	*10*
13C5 PFPeA	*58*		*25 - 150*	*10/16/18 08:36*	*10/25/18 04:26*	*10*
13C2 PFHxA	*62*		*25 - 150*	*10/16/18 08:36*	*10/25/18 04:26*	*10*
13C4 PFHpA	*81*		*25 - 150*	*10/16/18 08:36*	*10/25/18 04:26*	*10*
13C4 PFOA	*92*		*25 - 150*	*10/16/18 08:36*	*10/25/18 04:26*	*10*
13C5 PFNA	*91*		*25 - 150*	*10/16/18 08:36*	*10/25/18 04:26*	*10*
13C2 PFDA	*92*		*25 - 150*	*10/16/18 08:36*	*10/25/18 04:26*	*10*
13C2 PFUnA	*85*		*25 - 150*	*10/16/18 08:36*	*10/25/18 04:26*	*10*
13C2 PFDoA	*55*		*25 - 150*	*10/16/18 08:36*	*10/25/18 04:26*	*10*
13C2 PFTeDA	*15*	***	*25 - 150*	*10/16/18 08:36*	*10/25/18 04:26*	*10*
13C3 PFBS	*108*		*25 - 150*	*10/16/18 08:36*	*10/25/18 04:26*	*10*
18O2 PFHxS	*84*		*25 - 150*	*10/16/18 08:36*	*10/25/18 04:26*	*10*
13C4 PFOS	*91*		*25 - 150*	*10/16/18 08:36*	*10/25/18 04:26*	*10*
13C8 FOSA	*80*		*25 - 150*	*10/16/18 08:36*	*10/25/18 04:26*	*10*
d3-NMeFOSAA	*92*		*25 - 150*	*10/16/18 08:36*	*10/25/18 04:26*	*10*
d5-NEtFOSAA	*91*		*25 - 150*	*10/16/18 08:36*	*10/25/18 04:26*	*10*
M2-8:2 FTS	*132*		*25 - 150*	*10/16/18 08:36*	*10/25/18 04:26*	*10*

Client Sample ID: Leachate

Date Collected: 10/03/18 09:30
Date Received: 10/04/18 12:41

Lab Sample ID: 190-17646-1

Matrix: Water

Method: 537 (modified) - Fluorinated Alkyl Substances - DL

Analyte	Result	Qualifier	RL		Unit	D	Prepared	Analyzed	Dil Fac
6:2 FTS	<2100		2100		ng/L		10/16/18 08:36	10/25/18 04:19	100

Isotope Dilution	*%Recovery*	*Qualifier*	*Limits*	*Prepared*	*Analyzed*	*Dil Fac*
M2-6:2 FTS	*114*		*25 - 150*	*10/16/18 08:36*	*10/25/18 04:19*	*100*

Client: City of Cadillac Utilities
Project/Site: City of Cadillac

TestAmerica Job ID: 190-17646-1

Client Sample ID: Leachate (duplicate)

Lab Sample ID: 190-17646-2

Date Collected: 10/03/18 09:35
Matrix: Water
Date Received: 10/04/18 12:41

Method: 537 (modified) - Fluorinated Alkyl Substances

Analyte	Result	Qualifier	RL		Unit	D	Prepared	Analyzed	Dil Fac
Perfluoropentanoic acid (PFPeA)	**620**		18		ng/L		10/16/18 08:36	10/25/18 04:41	10
Perfluorohexanoic acid (PFHxA)	**1900**		18		ng/L		10/16/18 08:36	10/25/18 04:41	10
Perfluoroheptanoic acid (PFHpA)	**260**		18		ng/L		10/16/18 08:36	10/25/18 04:41	10
Perfluorooctanoic acid (PFOA)	**700**		18		ng/L		10/16/18 08:36	10/25/18 04:41	10
Perfluorononanoic acid (PFNA)	**18**		18		ng/L		10/16/18 08:36	10/25/18 04:41	10
Perfluorodecanoic acid (PFDA)	<18		18		ng/L		10/16/18 08:36	10/25/18 04:41	10
Perfluoroundecanoic acid (PFUnA)	<18		18		ng/L		10/16/18 08:36	10/25/18 04:41	10
Perfluorododecanoic acid (PFDoA)	<18		18		ng/L		10/16/18 08:36	10/25/18 04:41	10
Perfluorotridecanoic acid (PFTriA)	<18		18		ng/L		10/16/18 08:36	10/25/18 04:41	10
Perfluorotetradecanoic acid (PFTeA)	<18		18		ng/L		10/16/18 08:36	10/25/18 04:41	10
Perfluorobutanesulfonic acid (PFBS)	**1200**		18		ng/L		10/16/18 08:36	10/25/18 04:41	10
Perfluoropentanesulfonic acid (PFPeS)	**200**		18		ng/L		10/16/18 08:36	10/25/18 04:41	10
Perfluorohexanesulfonic acid (PFHxS)	**720**		18		ng/L		10/16/18 08:36	10/25/18 04:41	10
Perfluoroheptanesulfonic Acid (PFHpS)	<18		18		ng/L		10/16/18 08:36	10/25/18 04:41	10
Perfluorooctanesulfonic acid (PFOS)	**140**		18		ng/L		10/16/18 08:36	10/25/18 04:41	10
Perfluorononanesulfonic acid (PFNS)	<18		18		ng/L		10/16/18 08:36	10/25/18 04:41	10
Perfluorodecanesulfonic acid (PFDS)	<18		18		ng/L		10/16/18 08:36	10/25/18 04:41	10
Perfluorooctanesulfonamide (FOSA)	<18		18		ng/L		10/16/18 08:36	10/25/18 04:41	10
N-methylperfluorooctanesulfonamidoacetic acid (NMeFOSAA)	<180		180		ng/L		10/16/18 08:36	10/25/18 04:41	10
N-ethylperfluorooctanesulfonamidoacetic acid (NEtFOSAA)	<180		180		ng/L		10/16/18 08:36	10/25/18 04:41	10
4:2 FTS	<180		180		ng/L		10/16/18 08:36	10/25/18 04:41	10

Isotope Dilution	*%Recovery*	*Qualifier*	*Limits*	*Prepared*	*Analyzed*	*Dil Fac*
13C5 PFPeA	*53*		*25 - 150*	*10/16/18 08:36*	*10/25/18 04:41*	*10*
13C2 PFHxA	*60*		*25 - 150*	*10/16/18 08:36*	*10/25/18 04:41*	*10*
13C4 PFHpA	*79*		*25 - 150*	*10/16/18 08:36*	*10/25/18 04:41*	*10*
13C4 PFOA	*88*		*25 - 150*	*10/16/18 08:36*	*10/25/18 04:41*	*10*
13C5 PFNA	*91*		*25 - 150*	*10/16/18 08:36*	*10/25/18 04:41*	*10*
13C2 PFDA	*102*		*25 - 150*	*10/16/18 08:36*	*10/25/18 04:41*	*10*
13C2 PFUnA	*85*		*25 - 150*	*10/16/18 08:36*	*10/25/18 04:41*	*10*
13C2 PFDoA	*60*		*25 - 150*	*10/16/18 08:36*	*10/25/18 04:41*	*10*
13C2 PFTeDA	*14*	*	*25 - 150*	*10/16/18 08:36*	*10/25/18 04:41*	*10*
13C3 PFBS	*85*		*25 - 150*	*10/16/18 08:36*	*10/25/18 04:41*	*10*
18O2 PFHxS	*83*		*25 - 150*	*10/16/18 08:36*	*10/25/18 04:41*	*10*
13C4 PFOS	*92*		*25 - 150*	*10/16/18 08:36*	*10/25/18 04:41*	*10*
13C8 FOSA	*87*		*25 - 150*	*10/16/18 08:36*	*10/25/18 04:41*	*10*
d3-NMeFOSAA	*102*		*25 - 150*	*10/16/18 08:36*	*10/25/18 04:41*	*10*
d5-NEtFOSAA	*100*		*25 - 150*	*10/16/18 08:36*	*10/25/18 04:41*	*10*

Method: 537 (modified) - Fluorinated Alkyl Substances - DL

Analyte	Result	Qualifier	RL		Unit	D	Prepared	Analyzed	Dil Fac
Perfluorobutanoic acid (PFBA)	**840**		180		ng/L		10/16/18 08:36	10/25/18 04:34	100
6:2 FTS	<1800		1800		ng/L		10/16/18 08:36	10/25/18 04:34	100
8:2 FTS	<1800		1800		ng/L		10/16/18 08:36	10/25/18 04:34	100

Client Sample ID: Leachate (duplicate) — Lab Sample ID: 190-17646-2

Date Collected: 10/03/18 09:35 — **Matrix: Water**
Date Received: 10/04/18 12:41

Isotope Dilution	%Recovery	Qualifier	Limits	Prepared	Analyzed	Dil Fac
13C4 PFBA	64		25 - 150	10/16/18 08:36	10/25/18 04:34	100
M2-6:2 FTS	146		25 - 150	10/16/18 08:36	10/25/18 04:34	100
M2-8:2 FTS	94		25 - 150	10/16/18 08:36	10/25/18 04:34	100

Client Sample ID: Equipment Blank

Date Collected: 10/03/18 09:25
Date Received: 10/04/18 12:41

Lab Sample ID: 190-17646-3

Matrix: Water

Method: 537 (modified) - Fluorinated Alkyl Substances

Analyte	Result	Qualifier	RL		Unit	D	Prepared	Analyzed	Dil Fac
Perfluorobutanoic acid (PFBA)	<2.4		2.4		ng/L		10/16/18 08:36	10/21/18 03:39	1
Perfluoropentanoic acid (PFPeA)	<2.4		2.4		ng/L		10/16/18 08:36	10/21/18 03:39	1
Perfluorohexanoic acid (PFHxA)	<2.4		2.4		ng/L		10/16/18 08:36	10/21/18 03:39	1
Perfluoroheptanoic acid (PFHpA)	<2.4		2.4		ng/L		10/16/18 08:36	10/21/18 03:39	1
Perfluorooctanoic acid (PFOA)	<2.4		2.4		ng/L		10/16/18 08:36	10/21/18 03:39	1
Perfluorononanoic acid (PFNA)	<2.4		2.4		ng/L		10/16/18 08:36	10/21/18 03:39	1
Perfluorodecanoic acid (PFDA)	<2.4		2.4		ng/L		10/16/18 08:36	10/21/18 03:39	1
Perfluoroundecanoic acid (PFUnA)	<2.4		2.4		ng/L		10/16/18 08:36	10/21/18 03:39	1
Perfluorododecanoic acid (PFDoA)	<2.4		2.4		ng/L		10/16/18 08:36	10/21/18 03:39	1
Perfluorotridecanoic acid (PFTriA)	<2.4		2.4		ng/L		10/16/18 08:36	10/21/18 03:39	1
Perfluorotetradecanoic acid (PFTeA)	<2.4		2.4		ng/L		10/16/18 08:36	10/21/18 03:39	1
Perfluorobutanesulfonic acid (PFBS)	<2.4		2.4		ng/L		10/16/18 08:36	10/21/18 03:39	1
Perfluoropentanesulfonic acid (PFPeS)	<2.4		2.4		ng/L		10/16/18 08:36	10/21/18 03:39	1
Perfluorohexanesulfonic acid (PFHxS)	<2.4		2.4		ng/L		10/16/18 08:36	10/21/18 03:39	1
Perfluoroheptanesulfonic Acid (PFHpS)	<2.4		2.4		ng/L		10/16/18 08:36	10/21/18 03:39	1
Perfluorooctanesulfonic acid (PFOS)	<2.4		2.4		ng/L		10/16/18 08:36	10/21/18 03:39	1
Perfluorononanesulfonic acid (PFNS)	<2.4		2.4		ng/L		10/16/18 08:36	10/21/18 03:39	1
Perfluorodecanesulfonic acid (PFDS)	<2.4		2.4		ng/L		10/16/18 08:36	10/21/18 03:39	1
Perfluorooctanesulfonamide (FOSA)	<2.4		2.4		ng/L		10/16/18 08:36	10/21/18 03:39	1
N-methylperfluorooctanesulfonamidoacetic acid (NMeFOSAA)	<24		24		ng/L		10/16/18 08:36	10/21/18 03:39	1
N-ethylperfluorooctanesulfonamidoacetic acid (NEtFOSAA)	<24		24		ng/L		10/16/18 08:36	10/21/18 03:39	1
4:2 FTS	<24		24		ng/L		10/16/18 08:36	10/21/18 03:39	1
6:2 FTS	<24		24		ng/L		10/16/18 08:36	10/21/18 03:39	1
8:2 FTS	<24		24		ng/L		10/16/18 08:36	10/21/18 03:39	1

Isotope Dilution	*%Recovery*	*Qualifier*	*Limits*	*Prepared*	*Analyzed*	*Dil Fac*
13C4 PFBA	*95*		*25 - 150*	*10/16/18 08:36*	*10/21/18 03:39*	*1*
13C5 PFPeA	*96*		*25 - 150*	*10/16/18 08:36*	*10/21/18 03:39*	*1*
13C2 PFHxA	*92*		*25 - 150*	*10/16/18 08:36*	*10/21/18 03:39*	*1*
13C4 PFHpA	*99*		*25 - 150*	*10/16/18 08:36*	*10/21/18 03:39*	*1*
13C4 PFOA	*97*		*25 - 150*	*10/16/18 08:36*	*10/21/18 03:39*	*1*
13C5 PFNA	*97*		*25 - 150*	*10/16/18 08:36*	*10/21/18 03:39*	*1*
13C2 PFDA	*103*		*25 - 150*	*10/16/18 08:36*	*10/21/18 03:39*	*1*
13C2 PFUnA	*97*		*25 - 150*	*10/16/18 08:36*	*10/21/18 03:39*	*1*
13C2 PFDoA	*95*		*25 - 150*	*10/16/18 08:36*	*10/21/18 03:39*	*1*
13C2 PFTeDA	*94*		*25 - 150*	*10/16/18 08:36*	*10/21/18 03:39*	*1*
13C3 PFBS	*94*		*25 - 150*	*10/16/18 08:36*	*10/21/18 03:39*	*1*
18O2 PFHxS	*94*		*25 - 150*	*10/16/18 08:36*	*10/21/18 03:39*	*1*
13C4 PFOS	*99*		*25 - 150*	*10/16/18 08:36*	*10/21/18 03:39*	*1*
13C8 FOSA	*90*		*25 - 150*	*10/16/18 08:36*	*10/21/18 03:39*	*1*
d3-NMeFOSAA	*99*		*25 - 150*	*10/16/18 08:36*	*10/21/18 03:39*	*1*
d5-NEtFOSAA	*95*		*25 - 150*	*10/16/18 08:36*	*10/21/18 03:39*	*1*
M2-6:2 FTS	*105*		*25 - 150*	*10/16/18 08:36*	*10/21/18 03:39*	*1*
M2-8:2 FTS	*104*		*25 - 150*	*10/16/18 08:36*	*10/21/18 03:39*	*1*

Client Sample Results

Client: City of Cadillac Utilities
Project/Site: City of Cadillac

TestAmerica Job ID: 190-17646-1

Client Sample ID: Field Blank

Date Collected: 10/03/18 00:00
Date Received: 10/04/18 12:41

Lab Sample ID: 190-17646-4
Matrix: Water

Method: 537 (modified) - Fluorinated Alkyl Substances

Analyte	Result	Qualifier	RL		Unit	D	Prepared	Analyzed	Dil Fac
Perfluorobutanoic acid (PFBA)	<1.7		1.7		ng/L		10/16/18 08:36	10/21/18 03:46	1
Perfluoropentanoic acid (PFPeA)	<1.7		1.7		ng/L		10/16/18 08:36	10/21/18 03:46	1
Perfluorohexanoic acid (PFHxA)	<1.7		1.7		ng/L		10/16/18 08:36	10/21/18 03:46	1
Perfluoroheptanoic acid (PFHpA)	<1.7		1.7		ng/L		10/16/18 08:36	10/21/18 03:46	1
Perfluorooctanoic acid (PFOA)	<1.7		1.7		ng/L		10/16/18 08:36	10/21/18 03:46	1
Perfluorononanoic acid (PFNA)	<1.7		1.7		ng/L		10/16/18 08:36	10/21/18 03:46	1
Perfluorodecanoic acid (PFDA)	<1.7		1.7		ng/L		10/16/18 08:36	10/21/18 03:46	1
Perfluoroundecanoic acid (PFUnA)	<1.7		1.7		ng/L		10/16/18 08:36	10/21/18 03:46	1
Perfluorododecanoic acid (PFDoA)	<1.7		1.7		ng/L		10/16/18 08:36	10/21/18 03:46	1
Perfluorotridecanoic acid (PFTriA)	<1.7		1.7		ng/L		10/16/18 08:36	10/21/18 03:46	1
Perfluorotetradecanoic acid (PFTeA)	<1.7		1.7		ng/L		10/16/18 08:36	10/21/18 03:46	1
Perfluorobutanesulfonic acid (PFBS)	<1.7		1.7		ng/L		10/16/18 08:36	10/21/18 03:46	1
Perfluoropentanesulfonic acid (PFPeS)	<1.7		1.7		ng/L		10/16/18 08:36	10/21/18 03:46	1
Perfluorohexanesulfonic acid (PFHxS)	<1.7		1.7		ng/L		10/16/18 08:36	10/21/18 03:46	1
Perfluoroheptanesulfonic Acid (PFHpS)	<1.7		1.7		ng/L		10/16/18 08:36	10/21/18 03:46	1
Perfluorooctanesulfonic acid (PFOS)	<1.7		1.7		ng/L		10/16/18 08:36	10/21/18 03:46	1
Perfluorononanesulfonic acid (PFNS)	<1.7		1.7		ng/L		10/16/18 08:36	10/21/18 03:46	1
Perfluorodecanesulfonic acid (PFDS)	<1.7		1.7		ng/L		10/16/18 08:36	10/21/18 03:46	1
Perfluorooctanesulfonamide (FOSA)	<1.7		1.7		ng/L		10/16/18 08:36	10/21/18 03:46	1
N-methylperfluorooctanesulfonamidoacetic acid (NMeFOSAA)	<17		17		ng/L		10/16/18 08:36	10/21/18 03:46	1
N-ethylperfluorooctanesulfonamidoacetic acid (NEtFOSAA)	<17		17		ng/L		10/16/18 08:36	10/21/18 03:46	1
4:2 FTS	<17		17		ng/L		10/16/18 08:36	10/21/18 03:46	1
6:2 FTS	<17		17		ng/L		10/16/18 08:36	10/21/18 03:46	1
8:2 FTS	<17		17		ng/L		10/16/18 08:36	10/21/18 03:46	1

Isotope Dilution	*%Recovery*	*Qualifier*	*Limits*	*Prepared*	*Analyzed*	*Dil Fac*
13C4 PFBA	*98*		*25 - 150*	*10/16/18 08:36*	*10/21/18 03:46*	*1*
13C5 PFPeA	*101*		*25 - 150*	*10/16/18 08:36*	*10/21/18 03:46*	*1*
13C2 PFHxA	*99*		*25 - 150*	*10/16/18 08:36*	*10/21/18 03:46*	*1*
13C4 PFHpA	*95*		*25 - 150*	*10/16/18 08:36*	*10/21/18 03:46*	*1*
13C4 PFOA	*93*		*25 - 150*	*10/16/18 08:36*	*10/21/18 03:46*	*1*
13C5 PFNA	*98*		*25 - 150*	*10/16/18 08:36*	*10/21/18 03:46*	*1*
13C2 PFDA	*100*		*25 - 150*	*10/16/18 08:36*	*10/21/18 03:46*	*1*
13C2 PFUnA	*97*		*25 - 150*	*10/16/18 08:36*	*10/21/18 03:46*	*1*
13C2 PFDoA	*92*		*25 - 150*	*10/16/18 08:36*	*10/21/18 03:46*	*1*
13C2 PFTeDA	*100*		*25 - 150*	*10/16/18 08:36*	*10/21/18 03:46*	*1*
13C3 PFBS	*100*		*25 - 150*	*10/16/18 08:36*	*10/21/18 03:46*	*1*
18O2 PFHxS	*93*		*25 - 150*	*10/16/18 08:36*	*10/21/18 03:46*	*1*
13C4 PFOS	*100*		*25 - 150*	*10/16/18 08:36*	*10/21/18 03:46*	*1*
13C8 FOSA	*93*		*25 - 150*	*10/16/18 08:36*	*10/21/18 03:46*	*1*
d3-NMeFOSAA	*98*		*25 - 150*	*10/16/18 08:36*	*10/21/18 03:46*	*1*
d5-NEtFOSAA	*98*		*25 - 150*	*10/16/18 08:36*	*10/21/18 03:46*	*1*
M2-6:2 FTS	*96*		*25 - 150*	*10/16/18 08:36*	*10/21/18 03:46*	*1*
M2-8:2 FTS	*94*		*25 - 150*	*10/16/18 08:36*	*10/21/18 03:46*	*1*

Method: 537 (modified) - Fluorinated Alkyl Substances

Lab Sample ID: MB 320-252552/1-A
Matrix: Water
Analysis Batch: 253691

Client Sample ID: Method Blank
Prep Type: Total/NA
Prep Batch: 252552

Analyte	MB Result	MB Qualifier	RL		Unit	D	Prepared	Analyzed	Dil Fac
Perfluorobutanoic acid (PFBA)	<2.0		2.0		ng/L		10/16/18 08:36	10/21/18 01:25	1
Perfluoropentanoic acid (PFPeA)	<2.0		2.0		ng/L		10/16/18 08:36	10/21/18 01:25	1
Perfluorohexanoic acid (PFHxA)	<2.0		2.0		ng/L		10/16/18 08:36	10/21/18 01:25	1
Perfluoroheptanoic acid (PFHpA)	<2.0		2.0		ng/L		10/16/18 08:36	10/21/18 01:25	1
Perfluorooctanoic acid (PFOA)	<2.0		2.0		ng/L		10/16/18 08:36	10/21/18 01:25	1
Perfluorononanoic acid (PFNA)	<2.0		2.0		ng/L		10/16/18 08:36	10/21/18 01:25	1
Perfluorodecanoic acid (PFDA)	<2.0		2.0		ng/L		10/16/18 08:36	10/21/18 01:25	1
Perfluoroundecanoic acid (PFUnA)	<2.0		2.0		ng/L		10/16/18 08:36	10/21/18 01:25	1
Perfluorododecanoic acid (PFDoA)	<2.0		2.0		ng/L		10/16/18 08:36	10/21/18 01:25	1
Perfluorotridecanoic acid (PFTriA)	<2.0		2.0		ng/L		10/16/18 08:36	10/21/18 01:25	1
Perfluorotetradecanoic acid (PFTeA)	<2.0		2.0		ng/L		10/16/18 08:36	10/21/18 01:25	1
Perfluorobutanesulfonic acid (PFBS)	<2.0		2.0		ng/L		10/16/18 08:36	10/21/18 01:25	1
Perfluoropentanesulfonic acid (PFPeS)	<2.0		2.0		ng/L		10/16/18 08:36	10/21/18 01:25	1
Perfluorohexanesulfonic acid (PFHxS)	<2.0		2.0		ng/L		10/16/18 08:36	10/21/18 01:25	1
Perfluoroheptanesulfonic Acid (PFHpS)	<2.0		2.0		ng/L		10/16/18 08:36	10/21/18 01:25	1
Perfluorooctanesulfonic acid (PFOS)	<2.0		2.0		ng/L		10/16/18 08:36	10/21/18 01:25	1
Perfluorononanesulfonic acid (PFNS)	<2.0		2.0		ng/L		10/16/18 08:36	10/21/18 01:25	1
Perfluorodecanesulfonic acid (PFDS)	<2.0		2.0		ng/L		10/16/18 08:36	10/21/18 01:25	1
Perfluorooctanesulfonamide (FOSA)	<2.0		2.0		ng/L		10/16/18 08:36	10/21/18 01:25	1
N-methylperfluorooctanesulfonamidoacetic acid (NMeFOSAA)	<20		20		ng/L		10/16/18 08:36	10/21/18 01:25	1
N-ethylperfluorooctanesulfonamidoacetic acid (NEtFOSAA)	<20		20		ng/L		10/16/18 08:36	10/21/18 01:25	1
4:2 FTS	<20		20		ng/L		10/16/18 08:36	10/21/18 01:25	1
6:2 FTS	<20		20		ng/L		10/16/18 08:36	10/21/18 01:25	1
8:2 FTS	<20		20		ng/L		10/16/18 08:36	10/21/18 01:25	1

Isotope Dilution	*MB %Recovery*	*MB Qualifier*	*Limits*	*Prepared*	*Analyzed*	*Dil Fac*
13C4 PFBA	*85*		*25 - 150*	*10/16/18 08:36*	*10/21/18 01:25*	*1*
13C5 PFPeA	*89*		*25 - 150*	*10/16/18 08:36*	*10/21/18 01:25*	*1*
13C2 PFHxA	*87*		*25 - 150*	*10/16/18 08:36*	*10/21/18 01:25*	*1*
13C4 PFHpA	*95*		*25 - 150*	*10/16/18 08:36*	*10/21/18 01:25*	*1*
13C4 PFOA	*91*		*25 - 150*	*10/16/18 08:36*	*10/21/18 01:25*	*1*
13C5 PFNA	*92*		*25 - 150*	*10/16/18 08:36*	*10/21/18 01:25*	*1*
13C2 PFDA	*87*		*25 - 150*	*10/16/18 08:36*	*10/21/18 01:25*	*1*
13C2 PFUnA	*91*		*25 - 150*	*10/16/18 08:36*	*10/21/18 01:25*	*1*
13C2 PFDoA	*83*		*25 - 150*	*10/16/18 08:36*	*10/21/18 01:25*	*1*
13C2 PFTeDA	*86*		*25 - 150*	*10/16/18 08:36*	*10/21/18 01:25*	*1*
13C3 PFBS	*86*		*25 - 150*	*10/16/18 08:36*	*10/21/18 01:25*	*1*
18O2 PFHxS	*89*		*25 - 150*	*10/16/18 08:36*	*10/21/18 01:25*	*1*
13C4 PFOS	*90*		*25 - 150*	*10/16/18 08:36*	*10/21/18 01:25*	*1*
13C8 FOSA	*84*		*25 - 150*	*10/16/18 08:36*	*10/21/18 01:25*	*1*
d3-NMeFOSAA	*89*		*25 - 150*	*10/16/18 08:36*	*10/21/18 01:25*	*1*
d5-NEtFOSAA	*85*		*25 - 150*	*10/16/18 08:36*	*10/21/18 01:25*	*1*
M2-6:2 FTS	*88*		*25 - 150*	*10/16/18 08:36*	*10/21/18 01:25*	*1*
M2-8:2 FTS	*84*		*25 - 150*	*10/16/18 08:36*	*10/21/18 01:25*	*1*

Method: 537 (modified) - Fluorinated Alkyl Substances (Continued)

Lab Sample ID: LCS 320-252552/2-A
Matrix: Water
Analysis Batch: 253691

Client Sample ID: Lab Control Sample
Prep Type: Total/NA
Prep Batch: 252552

Analyte	Spike Added	LCS Result	LCS Qualifier	Unit	D	%Rec	%Rec. Limits
Perfluorobutanoic acid (PFBA)	40.0	41.3		ng/L		103	70 - 130
Perfluoropentanoic acid (PFPeA)	40.0	40.3		ng/L		101	66 - 126
Perfluorohexanoic acid (PFHxA)	40.0	38.2		ng/L		95	66 - 126
Perfluoroheptanoic acid (PFHpA)	40.0	38.2		ng/L		95	66 - 126
Perfluorooctanoic acid (PFOA)	40.0	37.6		ng/L		94	64 - 124
Perfluorononanoic acid (PFNA)	40.0	42.6		ng/L		106	68 - 128
Perfluorodecanoic acid (PFDA)	40.0	39.8		ng/L		100	69 - 129
Perfluoroundecanoic acid (PFUnA)	40.0	38.9		ng/L		97	60 - 120
Perfluorododecanoic acid (PFDoA)	40.0	38.7		ng/L		97	71 - 131
Perfluorotridecanoic acid (PFTriA)	40.0	42.3		ng/L		106	72 - 132
Perfluorotetradecanoic acid (PFTeA)	40.0	39.3		ng/L		98	68 - 128
Perfluorobutanesulfonic acid (PFBS)	35.4	32.7		ng/L		92	73 - 133
Perfluoropentanesulfonic acid (PFPeS)	37.5	40.5		ng/L		108	70 - 130
Perfluorohexanesulfonic acid (PFHxS)	36.4	32.8		ng/L		90	63 - 123
Perfluoroheptanesulfonic Acid (PFHpS)	38.1	37.8		ng/L		99	68 - 128
Perfluorooctanesulfonic acid (PFOS)	37.1	34.4		ng/L		93	67 - 127
Perfluorononanesulfonic acid (PFNS)	38.4	36.8		ng/L		96	70 - 130
Perfluorodecanesulfonic acid (PFDS)	38.6	36.8		ng/L		95	68 - 128
Perfluorooctanesulfonamide (FOSA)	40.0	38.0		ng/L		95	70 - 130
N-methylperfluorooctanesulfonamidoacetic acid (NMeFOSAA)	40.0	34.7		ng/L		87	67 - 127
N-ethylperfluorooctanesulfonamidoacetic acid (NEtFOSAA)	40.0	38.2		ng/L		96	65 - 125
4:2 FTS	37.4	36.3		ng/L		97	70 - 130
6:2 FTS	37.9	32.1		ng/L		85	66 - 126
8:2 FTS	38.3	36.7		ng/L		96	67 - 127

Isotope Dilution	*LCS %Recovery*	*LCS Qualifier*	*Limits*
13C4 PFBA	*90*		*25 - 150*
13C5 PFPeA	*90*		*25 - 150*
13C2 PFHxA	*94*		*25 - 150*
13C4 PFHpA	*93*		*25 - 150*
13C4 PFOA	*94*		*25 - 150*
13C5 PFNA	*90*		*25 - 150*
13C2 PFDA	*91*		*25 - 150*
13C2 PFUnA	*90*		*25 - 150*
13C2 PFDoA	*88*		*25 - 150*
13C2 PFTeDA	*96*		*25 - 150*
13C3 PFBS	*90*		*25 - 150*

Method: 537 (modified) - Fluorinated Alkyl Substances (Continued)

Lab Sample ID: LCS 320-252552/2-A
Matrix: Water
Analysis Batch: 253691

Client Sample ID: Lab Control Sample
Prep Type: Total/NA
Prep Batch: 252552

Isotope Dilution	LCS %Recovery	LCS Qualifier	Limits
18O2 PFHxS	96		25 - 150
13C4 PFOS	98		25 - 150
13C8 FOSA	88		25 - 150
d3-NMeFOSAA	95		25 - 150
d5-NEtFOSAA	95		25 - 150
M2-6:2 FTS	93		25 - 150
M2-8:2 FTS	93		25 - 150

Lab Sample ID: LCSD 320-252552/3-A
Matrix: Water
Analysis Batch: 253691

Client Sample ID: Lab Control Sample Dup
Prep Type: Total/NA
Prep Batch: 252552

Analyte	Spike Added	LCSD Result	LCSD Qualifier	Unit	D	%Rec	%Rec. Limits	RPD	RPD Limit
Perfluorobutanoic acid (PFBA)	40.0	41.3		ng/L		103	70 - 130	0	30
Perfluoropentanoic acid (PFPeA)	40.0	39.4		ng/L		98	66 - 126	2	30
Perfluorohexanoic acid (PFHxA)	40.0	41.5		ng/L		104	66 - 126	8	30
Perfluoroheptanoic acid (PFHpA)	40.0	39.0		ng/L		98	66 - 126	2	30
Perfluorooctanoic acid (PFOA)	40.0	42.3		ng/L		106	64 - 124	12	30
Perfluorononanoic acid (PFNA)	40.0	43.0		ng/L		108	68 - 128	1	30
Perfluorodecanoic acid (PFDA)	40.0	40.8		ng/L		102	69 - 129	2	30
Perfluoroundecanoic acid (PFUnA)	40.0	37.4		ng/L		94	60 - 120	4	30
Perfluorododecanoic acid (PFDoA)	40.0	41.0		ng/L		103	71 - 131	6	30
Perfluorotridecanoic acid (PFTriA)	40.0	43.6		ng/L		109	72 - 132	3	30
Perfluorotetradecanoic acid (PFTeA)	40.0	39.8		ng/L		100	68 - 128	1	30
Perfluorobutanesulfonic acid (PFBS)	35.4	34.0		ng/L		96	73 - 133	4	30
Perfluoropentanesulfonic acid (PFPeS)	37.5	40.4		ng/L		108	70 - 130	0	30
Perfluorohexanesulfonic acid (PFHxS)	36.4	33.5		ng/L		92	63 - 123	2	30
Perfluoroheptanesulfonic Acid (PFHpS)	38.1	37.3		ng/L		98	68 - 128	1	30
Perfluorooctanesulfonic acid (PFOS)	37.1	33.3		ng/L		90	67 - 127	3	30
Perfluorononanesulfonic acid (PFNS)	38.4	36.9		ng/L		96	70 - 130	0	30
Perfluorodecanesulfonic acid (PFDS)	38.6	36.7		ng/L		95	68 - 128	0	30
Perfluorooctanesulfonamide (FOSA)	40.0	38.9		ng/L		97	70 - 130	2	30
N-methylperfluorooctanesulfonamidoacetic acid (NMeFOSAA)	40.0	36.7		ng/L		92	67 - 127	6	30
N-ethylperfluorooctanesulfonamidoacetic acid (NEtFOSAA)	40.0	37.2		ng/L		93	65 - 125	3	30
4:2 FTS	37.4	36.2		ng/L		97	70 - 130	0	30
6:2 FTS	37.9	34.3		ng/L		90	66 - 126	7	30
8:2 FTS	38.3	39.3		ng/L		103	67 - 127	7	30

Isotope Dilution	*LCSD %Recovery*	*LCSD Qualifier*	*Limits*
13C4 PFBA	*97*		*25 - 150*
13C5 PFPeA	*98*		*25 - 150*
13C2 PFHxA	*93*		*25 - 150*
13C4 PFHpA	*97*		*25 - 150*
13C4 PFOA	*94*		*25 - 150*
13C5 PFNA	*95*		*25 - 150*
13C2 PFDA	*97*		*25 - 150*
13C2 PFUnA	*100*		*25 - 150*
13C2 PFDoA	*88*		*25 - 150*
13C2 PFTeDA	*95*		*25 - 150*
13C3 PFBS	*98*		*25 - 150*
18O2 PFHxS	*99*		*25 - 150*
13C4 PFOS	*105*		*25 - 150*
13C8 FOSA	*91*		*25 - 150*
d3-NMeFOSAA	*96*		*25 - 150*
d5-NEtFOSAA	*103*		*25 - 150*
M2-6:2 FTS	*96*		*25 - 150*
M2-8:2 FTS	*97*		*25 - 150*

1 2 3 4 5 6 7 8 9 10 11 12

Isotope Dilution Summary

Client: City of Cadillac Utilities

TestAmerica Job ID: 190-17646-1

Project/Site: City of Cadillac

Method: 537 (modified) - Fluorinated Alkyl Substances

Matrix: Water

Prep Type: Total/NA

Percent Isotope Dilution Recovery (Acceptance Limits)

Lab Sample ID	Client Sample ID	M262FTS (25-150)
190-17646-1 - DL	Leachate	114

Surrogate Legend

M262FTS = M2-6:2 FTS

Method: 537 (modified) - Fluorinated Alkyl Substances

Matrix: Water

Prep Type: Total/NA

Percent Isotope Dilution Recovery (Acceptance Limits)

Lab Sample ID	Client Sample ID	PFBA (25-150)	PFPeA (25-150)	PFHxA (25-150)	PFHpA (25-150)	PFOA (25-150)	PFNA (25-150)	PFDA (25-150)	PFUnA (25-150)
190-17646-1	Leachate	27	58	62	81	92	91	92	85
190-17646-2 - DL	Leachate (duplicate)	64							
190-17646-2	Leachate (duplicate)		53	60	79	88	91	102	85
190-17646-3	Equipment Blank	95	96	92	99	97	97	103	97
190-17646-4	Field Blank	98	101	99	95	93	98	100	97
LCS 320-252552/2-A	Lab Control Sample	90	90	94	93	94	90	91	90
LCSD 320-252552/3-A	Lab Control Sample Dup	97	98	93	97	94	95	97	100
MB 320-252552/1-A	Method Blank	85	89	87	95	91	92	87	91

Percent Isotope Dilution Recovery (Acceptance Limits)

Lab Sample ID	Client Sample ID	PFDoA (25-150)	PFTDA (25-150)	3C3-PFBS (25-150)	PFHxS (25-150)	PFOS (25-150)	PFOSA (25-150)	-NMeFOSA (25-150)	-NEtFOSA (25-150)
190-17646-1	Leachate	55	15 *	108	84	91	80	92	91
190-17646-2 - DL	Leachate (duplicate)								
190-17646-2	Leachate (duplicate)	60	14 *	85	83	92	87	102	100
190-17646-3	Equipment Blank	95	94	94	94	99	90	99	95
190-17646-4	Field Blank	92	100	100	93	100	93	98	98
LCS 320-252552/2-A	Lab Control Sample	88	96	90	96	98	88	95	95
LCSD 320-252552/3-A	Lab Control Sample Dup	88	95	98	99	105	91	96	103
MB 320-252552/1-A	Method Blank	83	86	86	89	90	84	89	85

Percent Isotope Dilution Recovery (Acceptance Limits)

Lab Sample ID	Client Sample ID	M262FTS (25-150)	M282FTS (25-150)
190-17646-1	Leachate		132
190-17646-2 - DL	Leachate (duplicate)	146	94
190-17646-2	Leachate (duplicate)		
190-17646-3	Equipment Blank	105	104
190-17646-4	Field Blank	96	94
LCS 320-252552/2-A	Lab Control Sample	93	93
LCSD 320-252552/3-A	Lab Control Sample Dup	96	97
MB 320-252552/1-A	Method Blank	88	84

Surrogate Legend

PFBA = 13C4 PFBA

PFPeA = 13C5 PFPeA

PFHxA = 13C2 PFHxA

PFHpA = 13C4 PFHpA

PFOA = 13C4 PFOA

PFNA = 13C5 PFNA

PFDA = 13C2 PFDA
PFUnA = 13C2 PFUnA
PFDoA = 13C2 PFDoA
PFTDA = 13C2 PFTeDA
13C3-PFBS = 13C3 PFBS
PFHxS = 18O2 PFHxS
PFOS = 13C4 PFOS
PFOSA = 13C8 FOSA
d3-NMeFOSAA = d3-NMeFOSAA
d5-NEtFOSAA = d5-NEtFOSAA
M262FTS = M2-6:2 FTS
M282FTS = M2-8:2 FTS

1 2 3 4 5 6 7 8 9 10 11 12

Qualifiers

LCMS

Qualifier	Qualifier Description
*	Isotope Dilution analyte is outside acceptance limits.

Glossary

Abbreviation	These commonly used abbreviations may or may not be present in this report.
¤	Listed under the "D" column to designate that the result is reported on a dry weight basis
%R	Percent Recovery
CFL	Contains Free Liquid
CNF	Contains No Free Liquid
DER	Duplicate Error Ratio (normalized absolute difference)
Dil Fac	Dilution Factor
DL	Detection Limit (DoD/DOE)
DL, RA, RE, IN	Indicates a Dilution, Re-analysis, Re-extraction, or additional Initial metals/anion analysis of the sample
DLC	Decision Level Concentration (Radiochemistry)
EDL	Estimated Detection Limit (Dioxin)
LOD	Limit of Detection (DoD/DOE)
LOQ	Limit of Quantitation (DoD/DOE)
MDA	Minimum Detectable Activity (Radiochemistry)
MDC	Minimum Detectable Concentration (Radiochemistry)
MDL	Method Detection Limit
ML	Minimum Level (Dioxin)
NC	Not Calculated
ND	Not Detected at the reporting limit (or MDL or EDL if shown)
PQL	Practical Quantitation Limit
QC	Quality Control
RER	Relative Error Ratio (Radiochemistry)
RL	Reporting Limit or Requested Limit (Radiochemistry)
RPD	Relative Percent Difference, a measure of the relative difference between two points
TEF	Toxicity Equivalent Factor (Dioxin)
TEQ	Toxicity Equivalent Quotient (Dioxin)

LCMS

Prep Batch: 252552

Lab Sample ID	Client Sample ID	Prep Type	Matrix	Method	Prep Batch
190-17646-1 - DL	Leachate	Total/NA	Water	3535	
190-17646-1	Leachate	Total/NA	Water	3535	
190-17646-2 - DL	Leachate (duplicate)	Total/NA	Water	3535	
190-17646-2	Leachate (duplicate)	Total/NA	Water	3535	
190-17646-3	Equipment Blank	Total/NA	Water	3535	
190-17646-4	Field Blank	Total/NA	Water	3535	
MB 320-252552/1-A	Method Blank	Total/NA	Water	3535	
LCS 320-252552/2-A	Lab Control Sample	Total/NA	Water	3535	
LCSD 320-252552/3-A	Lab Control Sample Dup	Total/NA	Water	3535	

Analysis Batch: 253691

Lab Sample ID	Client Sample ID	Prep Type	Matrix	Method	Prep Batch
190-17646-3	Equipment Blank	Total/NA	Water	537 (modified)	252552
190-17646-4	Field Blank	Total/NA	Water	537 (modified)	252552
MB 320-252552/1-A	Method Blank	Total/NA	Water	537 (modified)	252552
LCS 320-252552/2-A	Lab Control Sample	Total/NA	Water	537 (modified)	252552
LCSD 320-252552/3-A	Lab Control Sample Dup	Total/NA	Water	537 (modified)	252552

Analysis Batch: 254698

Lab Sample ID	Client Sample ID	Prep Type	Matrix	Method	Prep Batch
190-17646-1 - DL	Leachate	Total/NA	Water	537 (modified)	252552
190-17646-1	Leachate	Total/NA	Water	537 (modified)	252552
190-17646-2 - DL	Leachate (duplicate)	Total/NA	Water	537 (modified)	252552
190-17646-2	Leachate (duplicate)	Total/NA	Water	537 (modified)	252552

Lab Chronicle

Client: City of Cadillac Utilities
Project/Site: City of Cadillac

TestAmerica Job ID: 190-17646-1

Client Sample ID: Leachate

Lab Sample ID: 190-17646-1

Date Collected: 10/03/18 09:30
Date Received: 10/04/18 12:41

Matrix: Water

Prep Type	Batch Type	Batch Method	Run	Dilution Factor	Batch Number	Prepared or Analyzed	Analyst	Lab
Total/NA	Prep	3535	DL		252552	10/16/18 08:36	MYV	TAL SAC
Total/NA	Analysis	537 (modified)	DL	100	254698	10/25/18 04:19	JRB	TAL SAC
Total/NA	Prep	3535			252552	10/16/18 08:36	MYV	TAL SAC
Total/NA	Analysis	537 (modified)		10	254698	10/25/18 04:26	JRB	TAL SAC

Client Sample ID: Leachate (duplicate)

Lab Sample ID: 190-17646-2

Date Collected: 10/03/18 09:35
Date Received: 10/04/18 12:41

Matrix: Water

Prep Type	Batch Type	Batch Method	Run	Dilution Factor	Batch Number	Prepared or Analyzed	Analyst	Lab
Total/NA	Prep	3535	DL		252552	10/16/18 08:36	MYV	TAL SAC
Total/NA	Analysis	537 (modified)	DL	100	254698	10/25/18 04:34	JRB	TAL SAC
Total/NA	Prep	3535			252552	10/16/18 08:36	MYV	TAL SAC
Total/NA	Analysis	537 (modified)		10	254698	10/25/18 04:41	JRB	TAL SAC

Client Sample ID: Equipment Blank

Lab Sample ID: 190-17646-3

Date Collected: 10/03/18 09:25
Date Received: 10/04/18 12:41

Matrix: Water

Prep Type	Batch Type	Batch Method	Run	Dilution Factor	Batch Number	Prepared or Analyzed	Analyst	Lab
Total/NA	Prep	3535			252552	10/16/18 08:36	MYV	TAL SAC
Total/NA	Analysis	537 (modified)		1	253691	10/21/18 03:39	AAR	TAL SAC

Client Sample ID: Field Blank

Lab Sample ID: 190-17646-4

Date Collected: 10/03/18 00:00
Date Received: 10/04/18 12:41

Matrix: Water

Prep Type	Batch Type	Batch Method	Run	Dilution Factor	Batch Number	Prepared or Analyzed	Analyst	Lab
Total/NA	Prep	3535			252552	10/16/18 08:36	MYV	TAL SAC
Total/NA	Analysis	537 (modified)		1	253691	10/21/18 03:46	AAR	TAL SAC

Laboratory References:

TAL SAC = TestAmerica Sacramento, 880 Riverside Parkway, West Sacramento, CA 95605, TEL (916)373-5600

Analyst References:

Lab: TAL SAC

Batch Type: Prep

MYV = Mai Yee Vang

Batch Type: Analysis

AAR = Amani Royce

JRB = John Barnett

Accreditation/Certification Summary

Client: City of Cadillac Utilities
Project/Site: City of Cadillac

TestAmerica Job ID: 190-17646-1

Laboratory: TestAmerica Michigan

All accreditations/certifications held by this laboratory are listed. Not all accreditations/certifications are applicable to this report.

Authority	Program	EPA Region	Identification Number	Expiration Date
Michigan	State Program	5	57	05-05-20

Laboratory: TestAmerica Sacramento

All accreditations/certifications held by this laboratory are listed. Not all accreditations/certifications are applicable to this report.

Authority	Program	EPA Region	Identification Number	Expiration Date
Alaska (UST)	State Program	10	17-020	01-20-21
ANAB	DoD ELAP		L2468	01-20-21
Arizona	State Program	9	AZ0708	08-11-19
Arkansas DEQ	State Program	6	88-0691	06-17-19
California	State Program	9	2897	01-31-19
Colorado	State Program	8	CA00044	08-31-19
Connecticut	State Program	1	PH-0691	06-30-19
Florida	NELAP	4	E87570	06-30-19
Georgia	State Program	4	N/A	01-28-19
Hawaii	State Program	9	N/A	01-29-19
Illinois	NELAP	5	200060	03-17-19
Kansas	NELAP	7	E-10375	10-31-18 *
Louisiana	NELAP	6	30612	06-30-19
Maine	State Program	1	CA0004	04-14-20
Michigan	State Program	5	9947	01-31-20
Nevada	State Program	9	CA00044	07-31-19
New Hampshire	NELAP	1	2997	04-18-19
New Jersey	NELAP	2	CA005	06-30-19
New York	NELAP	2	11666	03-31-19
Oregon	NELAP	10	4040	01-29-19
Pennsylvania	NELAP	3	68-01272	03-31-19
Texas	NELAP	6	T104704399	05-31-19
US Fish & Wildlife	Federal		LE148388-0	07-31-19
USDA	Federal		P330-18-00239	01-17-21
USEPA UCMR	Federal	1	CA00044	12-31-20
Utah	NELAP	8	CA00044	02-28-19
Vermont	State Program	1	VT-4040	04-30-19
Virginia	NELAP	3	460278	03-14-19
Washington	State Program	10	C581	05-05-19
West Virginia (DW)	State Program	3	9930C	12-31-18
Wyoming	State Program	8	8TMS-L	01-28-19

* Accreditation/Certification renewal pending - accreditation/certification considered valid.

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Brighton, MI 48116
Phone (810) 229-2763 Fax (810) 229-0000

Chain of Custody Record

TestAmerica
THE LEADER IN ENVIRONMENTAL TESTING

Client Information

Client Contact: Cindy Tomaszewski	Sampler: Derek Goodrich	Lab PM: Schafer, Sue	Carrier Tracking No(s):	COC No: 190-21147-841.1
Company: City of Cadillac	Phone: 231-775-~~8~~2368	E-Mail: sue.schafer@testamericainc.com		Page: Page 1 of 1
Address: 1121 Plett Road	Due Date Requested:			Job #:
City: Cadillac	TAT Requested (days):			
State, Zip: MI, 49601				
Phone: 231-775-2368(Tel)	PO #: ~~Advance Payment Required~~ CK			
Email: lab@cadillac-mi.net	WO #:			
Project Name: City of Cadillac	Project #: 19000157			
Site: IPP MONITORING	SSOW#:			

Analysis Requested

Preservation Codes:
A - HCL; B - NaOH; C - Zn Acetate; D - Nitric Acid; E - NaHSO4; F - MeOH; G - Amchlor; H - Ascorbic Acid; I - Ice; J - DI Water; K - EDTA; L - EDA
M - Hexane; N - None; O - AsNaO2; P - Na2O4S; Q - Na2SO3; R - Na2S2O3; S - H2SO4; T - TSP Dodecahydrate; U - Acetone; V - MCAA; W - pH 4-5; Z - other (specify)
Other:

Sample Identification	Sample Date	Sample Time	Sample Type (C=comp, G=grab)	Matrix (W=water, S=solid, O=waste/oil, BT=Tissue, A=Air)	Field Filtered Sample (Yes or No)	Perform MS/MSD (Yes or No)	PFC_IDA - PFAS, Standard List (24 Analytes)	Total Number of containers	Special Instructions/Note:
			Preservation Code:				N		
LEACHATE	10-3-18	0930	G	Water	N	N	X	2	
LEACHATE (duplicate)	10-3-18	0935	G	Water	N	N	X	2	
EQUIPMENT BLANK	10-3-18	0925	G	Water	N	N		2	
FIELD BLANK	10-3-18	—	G	Water	N	N		2	

190-17646 Chain of Custody

Possible Hazard Identification
☐ Non-Hazard ☐ Flammable ☒ Skin Irritant ☐ Poison B ☐ Unknown ☐ Radiological

Deliverable Requested: I, II, III, IV, Other (specify)

Sample Disposal (A fee may be assessed if samples are retained longer than 1 month)
☐ Return To Client ☒ Disposal By Lab ☐ Archive For ________ Months

Special Instructions/QC Requirements:

Empty Kit Relinquished by:		Date:	Time:		Method of Shipment:	
Relinquished by: CA Tomaszewski	Date/Time: 10-3-2018 / 1400	Company: City of Cadillac	Received by: UPS Ground Ship	Date/Time: 10-3-2018 1400	Company: UPS	
Relinquished by: UPS	Date/Time:	Company	Received by: [signature]	Date/Time: 10/4/18 12:41	Company: TAL MI	
Relinquished by:	Date/Time:	Company	Received by:	Date/Time:	Company	
Custody Seals Intact: Δ Yes Δ No	Custody Seal No.:		Cooler Temperature(s) °C and Other Remarks:			

Ver: 08/04/2016

TestAmerica

Cooler/Sample Receipt

(AFTER HOURS receipt- complete gray areas
Place cooler in walk in. place this form in Receiving in-box Date Time rec'd ______ Initials ______

☐ MSDS or Known Hazard Information Supplied by Client
☐ Bottle stickers applied ☐ ELEMENT comment entered ☐ MSDS COC scanned emailed to EH&S
☐ Discrepancies Client ID City of Cadillac
☐ Short Hold Work Order # 190-17646
☐ Rush ☐24hr ☐2day ☐3day ☐5day ☐other ______
Receipt evaluation performed by - Initials AMY Date 10/4/18 Time 12:41

Method of Shipment:
☐ Walk-In Client ☐ TestAmerica Field/Courier
☐ Other Client/3rd Party Courier ______
☐ Fed Ex Tracking # ______
☒ UPS Tracking # 1ZE47 491 03 5470 1423
☐ Other Ground

Shipping Container Type:
☐ Cooler ☒ Box
☐ None ☒ Other Insulated Shipper

Packing Materials:
☒ Plastic Bags ☐ Foam
☐ Bubble Wrap ☐ Paper
☐ Packing Peanuts ☐ None
☐ Other ______

on inside box 216902/01

Custody Seals Intact:
☐ Yes ☐ No
☒ N/A (not used or required)

Cooling Materials:
☒ Ice (solid) ☐ Ice (Melted)
☐ Blue Ice ☐ None
☐ Other ______

Bacteriological Samples | Temp (oC) Corrected | Frozen yes no | Received within 2 hours yes no | Sample Flagged yes no

Receipt Temperatures ☐ Check if Additional Sheets Required

C.F. 0.0

Thermometer ID	Observed (°C)	Corrected (°C)	Temp Blank	Sample Temp	Received on same day sampled?	Acceptable?*	Cooler ID	Note Affected Samples if temperature not acceptable
140252483			☐	☐	☐Yes ☐No	☐Yes ☐No		
140252476	0.0	0.0	☐	☒	☐Yes ☐No	☒Yes ☐No		
			☐	☐	☐Yes ☐No	☐Yes ☐No		

* Receipt temperatures are considered acceptable if the samples are received on the same day they were collected & show signs that the cooling process has started. Temperature acceptance for most tests is ≤6.0°C, but not frozen. For additional information, please refer to SOP DT-SCA-004 *Sample Receipt and Login, Attachment 2 – Holding Times, Preservation and Container Requirements*

Receipt Questions**	Y	N	n/a	"No" answers require additional comment
COC present & TA receipt signature, date, & time properly documented?	✓			
Containers & labels in good condition? (unbroken, not leaking, appropriately filled, labels legible & attached)	✓			
Appropriate containers used & adequate volume provided?	✓			Preserved Bottles Checked with pH Strips* Yes No
Number of sample containers match COC?	✓			
Samples received within hold time?	✓			
Samples submitted for GRO and Volatiles analyses (8260, 624, 524) received without headspace?			✓	
Was a Trip Blank received with VOA samples?			✓	
Were the samples free of any questionable physical conformities? For example, field duplicates or multiple bottles of the same sample do not significantly vary in appearance (color, proportion of solids, etc.)	✓			
Were the COC, bottle labels, and all other items free of all other discrepancies or issues that would need to be addressed with the Project Manager and/or Client?	✓			

** May not be applicable if samples are not for compliance testing

* Excludes FOG, Volatiles TOC Vials

Client Contact Record
Contact via: ☐ Phone ☐ Email ☐ Other ______ Person Contacted: ______ Date/Time: ______
☐ Discrepancy allowance agreement is on record in the client project file.
Discussion/Resolution:

Any additional documentation and clarification from client must be noted in the narrative and/or scanned into the COC directory

Reviewed by PM Signature ______ Date 10/4/18

WI Page 1 of 1

WI No. DT-SCA-WI-001.10
effective 06/11/12

TestAmerica Michigan
10448 Citation Drive Suite 200
Brighton, MI 48116
Phone (810) 229-2763 Fax (810) 229-0000

Chain of Custody Record

TestAmerica
THE LEADER IN ENVIRONMENTAL TESTING

Client Information (Sub Contract Lab)	Sampler:	Lab PM: Schafer, Sue	Carrier Tracking No(s):	COC No: 190-20583.1
Client Contact: Shipping/Receiving	Phone:	E-Mail: sue.schafer@testamericainc.com	State of Origin: Michigan	Page: Page 1 of 1
Company: TestAmerica Laboratories, Inc.		Accreditations Required (See note):		Job #: 190-17646-1
Address: 880 Riverside Parkway,	Due Date Requested: 10/22/2018	Analysis Requested		Preservation Codes:
City: West Sacramento	TAT Requested (days):			
State, Zip: CA, 95605				
Phone: 916-373-5600(Tel) 916-372-1059(Fax)	PO #:			
Email:	WO #:			
Project Name: City of Cadillac	Project #: 19001884			
Site:	SSOW#:			Other:

Preservation Codes:
A - HCL; B - NaOH; C - Zn Acetate; D - Nitric Acid; E - NaHSO4; F - MeOH; G - Amchlor; H - Ascorbic Acid; I - Ice; J - DI Water; K - EDTA; L - EDA
M - Hexane; N - None; O - AsNaO2; P - Na2O4S; Q - Na2SO3; R - Na2S2O3; S - H2SO4; T - TSP Dodecahydrate; U - Acetone; V - MCAA; W - pH 4-5; Z - other (specify)

Sample Identification - Client ID (Lab ID)	Sample Date	Sample Time	Sample Type (C=comp, G=grab)	Matrix (W=water, S=solid, O=waste/oil, BT=Tissue, A=Air)	Field Filtered Sample (Yes or No)	Perform MS/MSD (Yes or No)	PFC_IDA/3535_PFC PFAS, Standard List (24 Analytes)	Total Number of containers	Special Instructions/Note:
			Preservation Code:						
Leachate (190-17646-1)	10/3/18	09:30 Eastern		Water			X	2	
Leachate (duplicate (190-17646-2)	10/3/18	09:35 Eastern		Water			X	2	
Equipment Blank (190-17646-3)	10/3/18	09:25 Eastern		Water			X	2	
Field Blank (190-17646-4)	10/3/18	Eastern		Water			X	2	

Note: Since laboratory accreditations are subject to change, TestAmerica Laboratories, Inc. places the ownership of method, analyte & accreditation compliance upon out subcontract laboratories. This sample shipment is forwarded under chain-of-custody. If the laboratory does not currently maintain accreditation in the State of Origin listed above for analysis/tests/matrix being analyzed, the samples must be shipped back to the TestAmerica laboratory or other instructions will be provided. Any changes to accreditation status should be brought to TestAmerica Laboratories, Inc. attention immediately. If all requested accreditations are current to date, return the signed Chain of Custody attesting to said complicance to TestAmerica Laboratories, Inc.

Possible Hazard Identification *Unconfirmed*		*Sample Disposal (A fee may be assessed if samples are retained longer than 1 month)* ☐ *Return To Client* ☐ *Disposal By Lab* ☐ *Archive For* ______ *Months*	
Deliverable Requested: I, II, III, IV, Other (specify)	Primary Deliverable Rank: 2	Special Instructions/QC Requirements:	
Empty Kit Relinquished by:	Date:	Time:	Method of Shipment:
Relinquished by: [signature]	Date/Time: 10/4/18 1410	Company: TAL	Received by: [signature] / Date/Time: 10-5-18 950 / Company: TA-Sac
Relinquished by:	Date/Time:	Company:	Received by: / Date/Time: / Company:
Relinquished by:	Date/Time:	Company:	Received by: / Date/Time: / Company:
Custody Seals Intact: ✗ Yes Δ No	Custody Seal No.: 216752	Cooler Temperature(s) °C and Other Remarks: 2.1	

Ver: 09/20/2016

ATTACHMENT NAME:

deq-tou-WRD-IPP_PFAS_Guidance-ScreeningEvaluation_620434_7.pdf

ATTACHMENT TYPE:

Adobe Portable Document Format (PDF) compound image

Recommended PFAS Screening & Evaluation Procedure for Industrial Pretreatment Programs (IPPs)

Consistent with our letter dated February 20, 2018, the Michigan Department of Environmental Quality (DEQ) Water Resources Division (WRD) provides the following information in response to requests by IPP professionals for guidance regarding the process of evaluating potential sources and determining probable sources of PFAS. PFAS are perfluoroalkyl and polyfluoroalkyl substances (also referred to as PFCs), especially the specific chemicals PFOS (perfluorooctane sulfonate) and PFOA (perfluorooctanoic acid).

Contents

1. Identify *Potential* Sources

Determine what *potential* sources of PFOS and PFOA are in your community that may discharge to your publicly owned treatment works (POTW). Do you have any of the following sewer users?

a. Platers/Metal Finishers
b. Paper and Packaging Manufacturers
c. Tanneries and Leather/Fabric/Carpet Treaters
d. Manufacturers of Parts with PTFE (polytetraflurorethylene, Teflon type) Coatings (i.e., Bearings)
e. Landfill Leachate
f. Centralized Waste Treaters
g. Contaminated Sites
h. Any other known or suspected sources of PFAS

If you have these sites in your community, but they do not discharge process wastewater, or infiltrate contaminated groundwater or storm water runoff to your sanitary sewer, they are not required to be covered by your efforts under the IPP PFAS Initiative. However, if you have knowledge of and/or concerns about such sites, please contact your District IPP staff to refer the issue for appropriate DEQ follow-up. An example of such a site would be a site contaminated by plating wastes that may contain PFAS but does not discharge to your sanitary sewer (but may discharge to waters of the state or a storm sewer).

2. Identify *Probable* Sources

Determine which of these *potential* sources is a *probable* source of PFOS and/or PFOA that may discharge to your POTW. Of the potential sources that you have identified in item 1, you will need to evaluate which currently use or have used PFAS-containing chemicals. You will need to interview industry/significant industrial user (SIU) staff and review Safety Data Sheets and Material Safety Data Sheets (SDS/MSDS) to find out about the industrial processes at each potential source and what chemicals have been used and/or generated there. Some industry staff may need to consult with their chemical suppliers to determine the content of the chemicals used at their sites. Please see the attached list of suggested questions organized by potential source that you may use as a guideline.

a. Platers/Metal Finishers: PFOS-containing chemicals were used by electroplaters as a demister/defoamer/surfactant to control air emissions of hexavalent chromium since mid-1990s. It is also a wetting agent that may have been used for other purposes. Even if used many years ago, historical use of PFOS-containing chemicals is still a potential source of PFOS to POTWs/surface waters due to the persistence of the chemical and typical practices at platers, who may not dump plating tanks or clean tanks and secondary containment pits. Platers in Michigan and Minnesota were found to have PFOS contamination in their wastewater years after they discontinued use of PFOS-containing chemicals.

TIP: Hard chrome and decorative chrome plating using hexavalent chromium are the most likely sources of PFOS-containing chemicals, although other types of plating may have also used these chemicals. Etch tanks where hexavalent chromium is used as part of the process often use demisters/defoamers/surfactants to control air emissions.

PFOS-containing chemicals (greater than 1%) were banned specifically from chrome *electroplating* tanks only since September 2015. Other uses were not banned, but the manufacture of PFOS and PFOA has been largely phased out in the US and Europe. We recommend asking about PFAS in general since some PFAS may be precursors or otherwise create PFOS or PFOA. We have limited information to date on specific products, but examples of demisters/defoamers/surfactants used at sites found to have PFOS include:

- ANKOR WETTING AGENT F, manufactured by Enthone (MSDS listed “fluorinated surfactant(s)” of 1-5%)
- Clepo Chrome Mist Control, manufactured by MacDermid Incorporated (MSDS lists ingredients as “proprietary”)
- Fumetrol™ 140 Mist Suppressant (Atotech USA, Rock Hill, SC). Available MSDS indicate that it contained “organic fluorosulfonate” between 1%-7% by weight.

Mist suppressants presumed to contain PFOS that were studied by the U.S. Environmental Protection Agency, Region V, in their PFOS Chromium Electroplater Study, September 2009, included the following products:

- Benchmark Benchbrite STX
- Benchmark CFS
- MacDermid Proquel B
- MacDermid Macuplex STR
- Plating Process Systems PMS-R
- Femetrol-140
- Brite Guard AF-1 fume control

We recommend looking at the purpose of the product to determine if it may contain PFAS. Products claiming to reduce surface tension, act as a wetting agent, eliminate misting, lower drag-out or act as a surfactant may contain PFAS. Those sold by US companies prior to 2015 may contain PFOS. If the content of the products is unclear or suspected to have PFAS (even “PFOS-free” products), consider this user a probable source. We believe that platers have used demisters/defoamers/surfactants that may contain PFAS for air emissions control since at least the mid-1990s, but we have heard that some platers may have used such chemicals since the 1960s.

b. Paper and Packaging Manufacturers: Some paper and packaging manufacturers use PFAS coatings or treatments for oil and moisture repellency. Many of these industries discharge directly to receiving streams, but if one discharges to you, contact them to find out if they are or have discharged PFAS-containing chemicals to your POTW.

c. Tanneries and Leather/Fabric/Carpet Treaters: Factory-applied oil, water and stain repellents have been known to cause contamination. Applications include protective clothing or outerwear, umbrellas, tents, sails, architectural materials, carpets, and upholstery.

> **TIP:** PFAS chemicals used include PTFE (polytetrafluoroethylene)—used in porous, “breathable” fabrics such as Gortex™.

d. Manufacturers of Parts with PTFE (polytetraflurorethylene, Teflon type) Coatings (i.e., Bearings): Our understanding is incomplete, but PFOA contamination in New York, New Hampshire, and Vermont appears to have been caused by a company that placed a non-stick coating on parts such as bearings and wires.

> **TIP:** PFAS chemicals used or generated in the process of creating PTFE (polytetrafluoroethylene) coatings may include PFOA.

e. Landfill Leachate: Most landfills will have some PFAS due to disposal of consumer products, but we are asking you to look for significant sources, which would include landfills receiving industrial wastes from metal finishers, leather/fabric treaters, other sources where there would be concentrated chemicals. Landfills that discharge leachate to your POTW, either by truck or sanitary sewer, should be evaluated.

> **TIP:** PFAS were developed in the 1940s, although uses changed over time. Fabric treatments became prevalent in the 1970s to present. We believe that platers have used demisters/defoamers/surfactants that may contain PFAS for air emissions control since at least the mid-1990s but we have heard that some platers may have used such chemicals since the 1960s.

If sources include those types listed above, the landfill should be considered a probable source of PFAS and should be monitored unless the landfill can show that their specific sources did not discharge PFAS.

The DEQ Waste Management and Radiological Protection Division (WMRPD) is developing a plan and schedule with industry partners to sample and report leachate data from landfills and provide that data to IPPs and the DEQ. Participating landfills will arrange for sampling their leachate and will follow DEQ recommended sampling protocols and SOP’s.

If landfills discharge their leachate to your POTW and one or more wish to participate in WMRPD’s program, you are encouraged to request an extension to the requirement to submit the landfill data with the Interim Report that is due by June 29, 2018. If you have determined that your landfills are not probable sources, you do not need to request an extension. WRD encourages you to contact your landfill customers and to cooperate with this effort. As stated in the April 18, 2018, letter, you have until May 8, 2018, to request an extension or alternative monitoring plan.

Please be aware that complete PFAS laboratory reports supplied through the WMRPD effort must be evaluated when it is submitted to you and reported to the DEQ WRD. In addition, any landfill leachate data collected should be obtained by POTWs for their review.

f. Centralized Waste Treaters (CWT) accepting certain wastes such as industrial process wastewater, groundwater cleanup wastewater, or landfill leachate may be sources of PFAS to your POTW.

> **TIP:** Look for platers, paper manufacturers, treated leather/fabrics, tannery wastes, chemical manufacturers, manufacturers using non-stick coatings, groundwater cleanups from these types of industries, and any other known sources of PFAS. Review the past 20 years since PFAS are heavy and may stick to the bottoms and sides of treatment tanks. If sources include those types listed, the CWT should be considered a *probable* source of PFAS and should be monitored unless the CWT can show that their specific sources did not discharge PFAS.

g. Contaminated Sites: Historical dumping of industrial wastes with PFAS, AFFF (aqueous film-forming foam for fire suppression) -- training sites especially, and manufacturing sites where PFAS were used (e.g. platers) may be a concern to POTWs if contaminated groundwater is discharged to POTWs through groundwater cleanup discharges or infiltration to the sanitary sewers. Many of these sites are more likely an issue for direct discharges to lakes and streams (also an issue for storm sewers for those who work on MS4 or storm sewer issues). If you have knowledge of and/or concerns about direct discharges to lakes and streams from contaminated sites, please contact your District IPP staff to refer the issue to the appropriate DEQ staff for follow-up.

h. Any other known or suspected sources of PFAS: If you know of other potential sources of PFAS, even those that may not be SIUs, you should determine whether they are probable sources, Survey questions are attached.

3. Develop a Monitoring Plan

Develop a monitoring plan to monitor the *probable* sources identified in item 2 above. Include the details of when and where you will sample, as well as how you will conduct sampling, transport your samples and analyze them at your chosen laboratory. The DEQ is developing PFAS sampling guidance and standard operating procedures, which will be posted online when they are available. If you want to sample locations within the collection system to capture multiple probable sources or request an extension for the initial source monitoring, please prepare an alternative plan for submittal to DEQ WRD by May 8, 2018. If you want to propose an alternative monitoring plan, contact your Regional IPP PFAS Specialist. All alternative monitoring plans or requests for extension for the interim report will be submitted through MiWaters via the IPP Unscheduled Submissions for Approval.

4. Work with Industry to Reduce and/or Eliminate these Pollutants

If you find significant sources of PFOS and/or PFOA through your source monitoring, you are required to work with the industry to reduce and/or eliminate these pollutants from their effluent. This may include:

- **Product substitution:** Stop using the fluorinated chemicals and hex chrome; use PFOS-free demisters (but that doesn't mean they are totally PFAS-free).
- **Operational controls:** Segregating waste streams for off-site disposal; using improved fume hoods.
- **Pretreatment:** Granular activated carbon (GAC) has been the most commonly used in full-scale water treatment applications. Other options are ion exchange resins and membrane filtration.
- **Clean-up:** Cleaning and properly disposing of waste from plating tanks, sump pits, pretreatment tanks, etc. Look at the overall housekeeping to determine where PFAS laden waste may be lurking.

Please note that the DEQ WRD does not expect POTWs that identify sources of PFAS to force any users out of their systems. Instead, we encourage POTWs to take a systematic approach, working with the DEQ and other partners to evaluate options for reducing or treating sources of PFAS and to use the IPP process to ensure that water quality and public health are protected.

5. Monitor Effluent

If there are potential impacts to your POTW (i.e. you have confirmed sources with greater than the applicable water quality standards for PFOS and/or PFOA [see Table 1 below]), you are required to monitor your POTW effluent for PFAS. If your POTW effluent sample results are above the Water Quality Standards (see Table 1 below), you will need to submit the results to the DEQ within 10 days of becoming aware of the results via the NPDES Unscheduled Permit Required Report in MiWaters. The DEQ will evaluate the results and will follow up with you as needed in writing.

Table 1 – Applicable Rule 57 Values

Chemicals	HNV* (non drinking water)	HNV (drinking water)
PFOS (ng/L)	12	11
PFOA (ng/L)	12,000	420

*HNV – Human Non-cancer Value

6. Reporting

An interim report summarizing your evaluation and the data collected (along with your documentation) is required to be submitted via MiWaters. Unless you have received an extension approval from the DEQ WRD, you are required to submit an interim report by June 29, 2018. A specific form and Schedule of Compliance will be available in MiWaters for the submission. In the interim report you will be asked to provide the following:

- A list of potential sources of PFOS and/or PFOA and your determination/justification regarding whether they are a probable source.
- Your monitoring plan outlining the steps taken to sample your probable sources.
- Your sampling results. This includes the contract lab reports and an Excel spreadsheet that the WRD will develop for you as part of the MiWaters form.
- If applicable, a list of Industrial Users found to be sources of PFOS and/or PFOA through sampling. You will need to provide any steps taken (or will be taken) on behalf of the POTW and IU to reduce and/or eliminate those pollutants in their effluent.
- If applicable, POTW effluent PFAS results, including contract lab reports and an Excel spreadsheet that the WRD will develop for you as part of the MiWaters form.
- If your POTW effluent results were higher than the WQS, when and how you notified DEQ WRD.

Unless no PFAS sources were found after the interim report submission, you must continue to work with known sources to reduce and/or eliminate PFAS entering the POTW. Depending on the information submitted in the interim report, the DEQ may ask you to do additional effluent monitoring, biosolids monitoring, and/or source monitoring. The DEQ will make these requests in writing.

Unless the DEQ sends a letter saying no further action is needed in response to the Interim Report, you must submit a Summary Report, which is due October 26, 2018. A specific form and Schedule of Compliance will be available in MiWaters for the submission. In the summary report you will be asked to provide the following:

- Any additional potential source(s) identified since the interim report was submitted and your determination if they are a probable source or not.

- If you sampled the additional probable sources, your sample results including contract lab reports and an updated Excel spreadsheet.
- An updated list of Industrial Users found to be sources of PFOS and/or PFOA through sampling.
- An update on the steps that have been taken (or will be taken) on behalf of the POTW and IU to reduce and/or eliminate those pollutants at the industries identified in the interim report and at the additional sources you identified.
- Any additional PFAS sampling results, including POTW effluent and/or biosolids, IU effluent for probable or confirmed sources (contract lab reports and an updated Excel spreadsheet).

Additional efforts may be needed at some POTWs. Additional requirements will be addressed in writing by DEQ WRD on a case-by-case basis.

Suggested IU Survey Questions for Per- and Polyfluoroalkyl Substances (PFAS) Identification

Please review the list of potential PFAS sources below. If you currently or have historically discharged wastewater into the sanitary sewers from any of the manufacturing operations or activities listed, please complete the corresponding section identified and return it to (POTW ADDRESS) within 10 days.

- Electroplaters of metals or plastics, metal finishers, circuit board manufacturers – *Complete Section 1*
- Paper and packaging manufacturers – *Complete Section 2*
- Leather, textile, fabric and/or carpet treaters, leather tanneries – *Complete Section 3*
- Manufacturers of parts with polytetraflurorethylene (PFTE) coatings such as bearings, wires, etc. – *Complete Section 4*
- Landfills with leachate discharges – *Complete Section 5*
- Centralized Waste Treaters – *Complete Section 6*
- Industry sites with soil or groundwater contamination including those where aqueous film forming foam (AFFF) was used (e.g. fires, firefighter training, equipment testing) – *Complete Section 7*
- Industries using PFAS in other processes or operations – *Complete Section 8*

Section 1 - Electroplaters/metal finishers/circuit board manufacturers

PFAS-containing chemicals, specifically those containing perfluoroctane sulfonate (PFOS) were used by electroplaters as a demister/defoamer/surfactant to control air emissions of hexavalent chromium beginning in the mid-1990s. While hard chrome and decorative chrome platers using hexavalent chrome are the most likely sources, PFAS have also been found in wetting agents and other plating chemicals involving other metals and plastics. Even if used many years ago, PFAS-containing chemicals may persist in plating tanks, etch tanks, sumps, air emission control systems and secondary containment pits. Some chemicals identified as PFOS-free may still contain PFAS. We are still learning about the behavior of these chemicals, and there are concerns that chemical changes may occur in plating and etch baths. Platers in Michigan and Minnesota were found to have PFOS contamination in their wastewater years after they discontinued use of PFOS-containing chemicals.

i. What types of plating are currently performed at your facility?

ii. What types of plating were previously performed at your facility? Please summarize the plating activities over the last 20 years if possible.

iii. Are or were demisters/defoamers/surfactants used to control air emissions or as wetting agents for any plating or etch tanks? If so, what are the names of the chemicals and amounts and concentrations used? Please provide the SDS/MSDS sheets for these chemicals.

iv. Are any other chemicals used in the plating processes known to contain PFAS or PFOS? If so, what are the names of the chemicals and amounts and concentrations used? Please provide the SDS/MSDS sheets for these chemicals and a schematic or flow diagram that shows where each chemical is used.

v. Are there any other fluorinated chemicals used (look for “fluoro” in the SDS/MSDS chemical listing or product name, e.g., “fluorinated surfactant(s)” or “organic fluorosulfonate”)? If so, what are the names of the chemicals and amounts and concentrations used? Please provide the SDS/MSDS sheets for these chemicals and a description of where they are used in your process.

Section 2 – Paper and Packaging Manufacturers

Paper and packaging manufacturers: Some paper and packaging manufacturers use PFAS coatings or treatments for oil and moisture repellency.

i. Do you currently or have you in the past treated paper or packaging for oil- and/or water-repellency? In answering this question, please review your activities over the last 20 years if possible. Discuss the treatments used, time periods, percentage products treated, types of products treated or number of lines.

ii. Do any chemicals used to coat or treat the paper/packaging contain PFAS (look for “fluoro” in the SDS/MSDS chemical listing or product name, e.g.,“fluorinated surfactant(s)” or “organic fluorosulfonate” ? If so, what are/were the names of the chemicals and amounts and concentrations used? Please provide the SDS/MSDS sheets for these chemicals.

iii. Are there any other fluorinated chemicals used (look for “fluoro” in the SDS/MSDS chemical listing or product name, e.g., “fluorinated surfactant(s)” or “organic fluorosulfonate”) in your process? If so, what are the names of the chemicals and amounts and concentrations used? Please provide the SDS/MSDS sheets for these chemicals.

iv. Do you use recycled paper in your process? What is the source of the recycled paper and what percentage of recycled paper is used in the final product?

Section 3 – Leather, Textile, Fabric and/or Carpet Treaters, Leather Tanneries

Factory-applied repellents for stain, oil- and/or water have been known to contain PFAS. Applications include protective coatings on leather, clothing or outerwear, umbrellas, tents, sails, architectural materials, carpets, and upholstery.

i. Do you currently or have you in the past applied treatment to leather, fabrics, textiles or carpet for stain, oil- and/or water-repellency? In answering this question, please review your activities over the last 20 years if possible. Discuss the treatments used, time periods, percentage products treated, type products treated or number of lines.

ii. Do any of the chemicals used for treating contain PFAS (look for "fluoro" in the SDS/MSDS chemical listing or product name, e.g., "fluorinated surfactant(s)" or "organic fluorosulfonate")? If so, what are the names of the chemicals and amounts and concentrations used? Please provide the SDS/MSDS sheets for these chemicals.

iii. Are there any other fluorinated chemicals used in the treatments (look for "fluoro" in the SDS/MSDS chemical listing or product name, e.g., "fluorinated surfactant(s)" or "organic fluorosulfonate")? If so, what are the names of the chemicals and amounts and concentrations used? Please provide the SDS/MSDS sheets for these chemicals.

Section 4 – Manufacturers of parts with polytetrafluororethylene (PTFE) coatings

A PFAS containing chemical, perfluorooctanoic acid (PFOA), is used in the process of making PTFE, which is a form of Teflon. Our understanding is incomplete, but PFOA contamination in New York, New Hampshire, and Vermont appears to have been caused by a company that placed a non-stick coating on parts such as bearings and wires.

i. Do you currently or have you in the past coated parts with PTFE or similar non-stick coatings? Describe all waste streams from the process.

ii. What are the names of the chemicals used and/or created in the coating process? Please provide the amounts and concentrations used? Please provide the SDS/MSDS sheets for chemicals used. In answering this question, please review your manufacturing operations over the last 20 years if possible.

iii. What chemical byproducts are generated in the coating process?

iv. Are there any other PFAS or fluorinated chemicals used (look for "fluoro" in the SDS/MSDS chemical listing or product name, e.g., "fluorinated surfactant(s)" or "organic fluorosulfonate") or generated? If so, what are the names of the chemicals and amounts and concentrations used or generated? Please provide the SDS/MSDS sheets for these chemicals.

Section 5 – Landfills that Discharge Leachate

Most landfill leachate will have some PFAS due to disposal of consumer products, but we are currently interested in landfills that have received PFAS laden industrial wastes such as sludge from metal finishers, leather tanneries and/or similar sources.

i. Is your landfill still accepting wastes for disposal? If not, when was it closed?

ii. Do you currently or have you in the past accepted industrial wastes? Approximately what percentage of your wastes would be considered industrial?

iii. Were wastes from any of the following manufacturing operations accepted: Sludge from electroplaters/metal finishers, waste from coated/treated paper manufacturers, treated leather/fabric wastes, tannery wastes, wastes from PFAS chemical manufacturers, wastes from manufacturers using non-stick coatings, municipal biosolids known to be impacted by PFAS, any other known sources of PFAS? Please provide types and estimated amounts or percentage wastes received.

iv. Have you analyzed your leachate for PFAS? If so, please provide the results.

Section 6 – Centralized Waste Treaters (CWTs)

CWTs accepting certain wastes may be sources of PFAS depending on the types of wastewater they receive for treatment.

i. Do you currently or have you in the past accepted wastewater for treatment from electroplaters, metal finishers, coated/treated paper or packaging manufacturers, treated leather/fabrics manufacturers, leather tanneries, chemical manufacturers involved in PFAS production, manufacturers applying non-stick coatings such as PFTE or Teflon, firefighting foam from training facilities, spills or equipment testing, groundwater remediation of the aforementioned types of industries, storm water from the aforementioned types of industries and/or any other known sources of PFAS?

ii. Have you analyzed your wastewater for PFAS? If so, please provide the results.

Section 7 – Industry sites with soil or groundwater contamination including those where aqueous film forming foam (AFFF) was used:

If your industry or facility has soil or groundwater contamination due to releases of industrial wastes or the use of AFFF (Class B) firefighting foam due to fires or firefighter training that discharges or infiltrates into the sanitary sewers may be a concern.

i. Do you currently have contamination of soil or groundwater due to releases from electroplating/metal finishing processes, the coating/treatment of paper or packaging products, textile, leather or fabric treating, leather tanning operations, the manufacturing of PTFE coatings or other PFAS sources? Please describe below.

ii. Has your facility had a fire in which AFFF (Class B) foam was utilized, or has firefighter training occurred on your site using AFFF foam? Please describe below.

iii. Have you analyzed your groundwater for PFAS? If so, please provide the results.

iv. Do you have a groundwater cleanup or investigation? Please describe.

Section 8 – Industries using PFAS in other processes and operations.

If you are aware of the use of chemicals containing PFAS in your processes or operations please answer the questions below.

i. What are the names of the chemicals containing PFAS? Please provide the amounts and concentrations used? Please provide the SDS/MSDS sheets for the PFAS containing chemicals. In answering this question, please review your manufacturing operations over the last 20 years if possible.

ii. Are there any other fluorinated chemicals used (look for "fluoro" in the SDS/MSDS chemical listing or product name, e.g., "fluorinated surfactant(s)" or "organic fluorosulfonate")? If so, what are the names of the chemicals and amounts and concentrations used? Please provide the SDS/MSDS sheets for these chemicals.